MIL-O-36134C
18 July 1978
SUPERSEDING
MIL-O-36134B
16 April 1970

MILITARY SPECIFICATION

OTOSCOPE, BATTERY TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a battery-type otoscope suitable for use in examination of, and surgery within, the ear.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-B-18	- Battery, Dry.
MIL-P-116	- Preservation, Methods of.
MIL-M-7298	- Manuals, Technical, Commercial Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENT

3.1 Material. Glass parts shall be fabricated from high quality optical glass. All materials shall be of the highest quality throughout, and suitable in all respects for the purpose intended.

3.2 Design and construction. The otoscope shall consist essentially of the following:

a) Otoscope head with lamp.
b) Battery handle.
c) Three (3) aural specula: large, medium and small.
d) One (1) nasal speculum.
e) Two (2) replacement lamps.
f) Carrying case.

3.2.1 Otoscope head. The otoscope head shall contain condensing lens, lamp housing, magnifier, base connector, and specula. The head shall provide intense uniform light properly concentrated and focused just beyond the end of the speculum in such a manner as to minimize direct glare, shadows and back reflections. The axial illumination at a point located at the small end of the longest speculum shall not be less than 210 lumens per square foot (psf) when the lamp is operated at not more than 2.70 volts and tested as specified in 4.4.1. The optical system shall be such that it requires no adjustment of the lamp in order to obtain a clear undistorted view and uniform light density. The otoscope head shall be of the detachable, open head, dual purpose type, suitable for use in examination of, and for surgery inside the ear. The supporting member of the otoscope head shall be fabricated of a metal suitable for the intended purpose. The speculum holder shall be provided with a screw-mount, bayonet-mount, or friction-mount for securely mounting the specula in position during use. The otoscope head shall be so designed as to provide an unobstructed path and maximum working area for the insertion and manipulation of surgical instruments without significant reduction of light intensity at the speculum tip. To facilitate ready attachment and removal, the otoscope head shall be equipped with a spring-loaded bayonet-type connector with indexing lock mechanism.

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3.2.1.1 Lamp housing. The lamp housing shall be so designed as to provide ready accessibility for rapid replacement of the lamp. The lamp housing shall be designed for use with the lamp specified in 3.2.1.2. The housing shall be designed to insure proper positioning of the lamp in respect to the optical system. The lamp housing shall be fitted with a condensing lens. If a lamp of the pre-focused type is used the condensing lens shall not be required.

3.2.1.2 Lamp. The lamp shall be rated at a maximum of 0.4 amperes at 2.5 volts. The lamp shall be prefocused and precentered. The lamp shall contain a close-coiled tungsten filament capable of meeting the illumination requirements specified in 3.2.1. The lamp shall have a minimum life of 9 hours when tested as specified in 4.4.2.

3.2.1.3 Magnifier. The magnifier shall be permanently attached to the base connector by means of a journal stud. The magnifier shall be spring loaded to hold its adjustment. The magnifier shall be held on the stud by means of an austenitic stainless steel screw, and shall be capable of being moved out of the line of vision to permit oblique observation, observation without magnification, and the insertion of surgical instruments. The magnifying lens shall furnish a magnified view of at least 2 diameters.

3.2.1.4 Specula. The aural and nasal specula shall be fabricated of nylon or polypropylene. The interior surface shall be finished to be nonreflecting to prevent interfering glare to the operator. As an alternate, the plastic specula shall have a corrosion-resistant metal attachment band securely affixed to the proximal end for use in attaching the specula to the otoscope head. The distal ends of the specula shall have rounded edges, free of sharpness. The specula shall be designed to permit ready attachment to, or removal from, the otoscope head. The specula shall not be distorted, the finish shall not be marred, nor shall the metal attachment band, when supplied, loosen, become detached, or show signs of corrosion, when subjected to the boil test specified in 4.4.3. The specula shall not chip, fracture, become distorted, nor shall the attachment band, when supplied, loosen, or become detached, when subjected to the drop test specified in 4.4.4. There shall be no discoloration or dulling of the surface of the specula, or to the attachment band, when supplied, when subjected to the alcohol immersion test specified in 4.4.5.

3.2.1.5 Battery handle. The battery handle shall be cylindrical in shape and of a size to properly hold two BA-30 dry cells per MIL-B-18. The handle shall be fabricated of a suitable metal. One end of the handle shall be properly designed for a simple and rapid attachment of the otoscope head in a secure manner. The metal connector of the handle, for receiving the otoscope head shall be spring-loaded, bayonet-type, having an indexing lock mechanism. The connector assembly shall be provided with a stainless steel stud for supporting the head in the mounted position. The other end of the battery handle shall be provided with a suitable metal closure cap attachable to the handle by means of a bayonet-type or threaded fitting. If a threaded fitting is used, only machine cut threads shall be acceptable. The inside of the closure cap shall be equipped with a spring designed to hold the batteries securely in position. The battery handle shall be provided with a conveniently located rheostat, which shall effectively control the intensity of the light. The rheostat shall be easily operated and accessible with the otoscope head mounted to the battery handle. The rheostat shall be equipped with a locking mechanism to prevent the instruments from "turning-on" when not in use.

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3.2.2 Carrying case. The carrying case shall be molded of a durable, resilient rubber base material or plastic, washable with soap and water. There shall be a form-fitting molded insert in the bottom of the case to accommodate all the components (a thru e) specified in 3.2, and prevent their shifting and damage during carrying and storage. The case shall be provided with a continuous hinge or two suitable hinges, located one near each end of the case. The case shall be equipped with a suitable latch, or other suitable means of closure. As an alternate, the top and bottom sections of the carrying case shall have metal trimmed edges.

3.3 Performance.

3.3.1 Strength of otoscope. There shall be no misalignment of the optical system, or permanent bending or breakage of the instrument when tested as specified in 4.4.6.

3.3.2 Drop test. There shall be no damage to the carrying case or its contents, other than possible filament displacement or filament breakage of the lamps that are installed in the otoscope head when tested as specified in 4.4.7. However, after the test, a replacement bulb, removed from its compartment in the carrying case, and installed in the otoscope head, shall incandesce and provide a proper focus.

3.4 Finish.

3.4.1 Otoscope head. All exposed surfaces of the otoscope head shall be finished in non-gloss or semi-gloss, black or gray enamel, or shall be plated or treated to resist corrosion. If enamel is used, the enamel shall be synthetic resin-type, or flexible baked-on enamel.

3.4.2 Battery handle. The external surfaces of the battery case shall be chromium plated or covered with a seamless plastic that will not chip, flake, or peel during normal use.

3.5 Hardware. All hardware, such as screws, nuts, and washers shall be corrosion-resistant steel, or shall be suitably finished or plated to resist corrosion.

3.6 Identification marking. The otoscope head, the battery handle and carrying case shall be permanently and legibly marked with the name or registered trade mark of the contractor. In addition, the carrying case shall also be marked with the manufacturer's model number of the set, if such number is used. The markings on the carrying case may be accomplished by the use of an adhesive backed aluminum plate. The markings on the carrying case shall be either on the outside or the inside.

3.7 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the

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contract, the contractor shall furnish to the Contracting Officer two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review, based on a maximum period of 30 days for such review and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract. In addition, the Contractor shall furnish 15 copies of the approved technical manual to the Contracting Officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the Contracting Officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

3.8 Workmanship. Workmanship shall be first class throughout. The otoscope shall be free from defects which detract from its appearance or may impair its serviceability. All burrs and sharp edges shall be removed from the item.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be one of each component (i.e., one otoscope head, one battery handle, the specula, one lamp, one case).

Table I. Sampling for examination.

Type defect	Inspection level	AQL (percent defective)
Major A	S-2	1.0
Major B	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for sampling purposes shall be one of each component (see 4.2.1).

TABLE II. Sampling for tests.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Otoscope head	Axial illumination	3.2.1	4.4.1	S-1	1.5
Lamp	Lamp life	3.2.1.2	4.4.2	S-1	1.5
Speculum	Boil test	3.2.1.4	4.4.3	S-1	1.5
	Drop test	3.2.1.4	4.4.4	S-1	2.5
	Alcohol immersion	3.2.1.4	4.4.5	S-1	2.5
Otoscope	Strength	3.3.1	4.4.6	S-1	2.5
Set	Drop test	3.3.2	4.4.7	S-1	2.5

4.3 Examination.

4.3.1 Otoscope. The otoscope shall be examined to determine compliance with all the requirements contained herein. Examination shall be conducted as specified in table I and table III and in accordance with MIL-STD-105. Inspection, however, shall not be restricted to defects listed in table III.

TABLE III. Classification of defects.

Categories	Defects
	Otoscope head
Major A:	
101	Clear, undistorted, view and uniform light intensity not obtained without adjustment of the lamp.
102	Magnifier distorted, scratched, chipped, not free of bubbles or striae, or otherwise defective so that the image is distorted, not clear and sharp.

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TABLE III. Classification of defects. (Cont'd)

Categories	Defects
Major B:	
151	Unit inoperative (i.e., otoscope head with battery handle).
152	Does not provide intense uniform light concentrated and focused as specified, resulting in direct glare, shadows, and back reflections.
153	Illuminated spot not uniform, not free of distortions.
154	Speculum holder does not hold specula securely.
155	Magnifier does not magnify at least 2 diameters.
156	Interior of specula not nonreflecting.
157	Distal ends of specula not rounded, not free of sharpness.
158	One or more specula missing.
159	Otoscope head not easily and smoothly attachable to, or detachable from, the battery handle.
160	Exposed parts not smooth and free of sharpness, burrs, or flash.
161	Finish is porous, peeling, or incomplete.
162	Part or parts missing.
Minor:	
201	Contains scratches, chips, or discoloration.
202	Identification marking missing or not legible.
203	Service date missing or not complete.
204	Replacement lamp missing or inoperative.
	Battery handle
Major A:	
101	Not equipped with a rheostat.
102	Rheostat not operative, or difficult to operate.
103	Not cylindrical and of a size to hold two of the specified batteries.
104	Not fabricated of metal.
105	Finish or covering not as specified.
106	Not provided with bayonet-type or thread type closure cap with spring for securely holding batteries in place.
107	Threads not cut threads.
108	Connector for securing head not spring-loaded bayonet-type with an indexing lock mechanism.
109	Head cannot be easily removed and replaced on battery handle.
110	Battery handle supplied with batteries.
111	Identification marking missing or illegible.
	Carrying case
Major B:	
101	Not fabricated as specified.
Minor:	
201	Contains scratches, chips, or discoloration.
202	Identification marking missing or illegible.

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4.3.2 Packaging inspection. The inspection of the unit package, packing, and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Axial illumination. The axial illumination shall be determined with the use of a Weston Model 614 Foot-Candle Meter, or an equivalent apparatus. The otoscope shall meet the requirements specified in 3.2.1.

4.4.2 Lamp life. The lamp shall be supplied with 2.5 ± 0.125 volts, direct current and maintained for 9 hours. The lamp shall meet the requirements of 3.2.1.2.

4.4.3 Boil test. Bring a beaker of distilled water to boiling temperature. Place a specula in the boiling water for a period of 16 ± 1 minutes. Remove the specula from the boiling water, and allow to air dry for a period of 10 ± 1 minutes. Repeat above test 4 times. The specula shall meet the requirements specified in 3.2.1.4.

4.4.4 Drop test. The specula shall be dropped upon a cement floor from a height of 5 ± 1/2 feet. The specula shall meet the requirements specified in 3.2.1.4.

4.4.5 Alcohol immersion. The specula shall be completely immersed in 95 percent ethyl alcohol for a period of 30 minutes. After removal from the alcohol, rinse thoroughly in distilled water and dry. The specula shall meet the requirements specified in 3.2.1.4.

4.4.6 Strength of otoscope head. The otoscope head shall be assembled to the battery handle. The battery handle shall be securely clamped, in a horizontal position, by a suitable means, with the speculum holder in the upward position. A weight of sufficient size to afford a 24 inch-pound bending moment at the base edge of the head connector shall be applied to the otoscope head. Maintain the load for 1 minute. The otoscope head shall meet the requirements specified in 3.3.1.

4.4.7 Drop test for set. The carrying case, with all components, including spare lamps, shall be dropped from a height of 5 ± 1/2 feet onto a concrete floor. There shall be five drops using random case positions. After the five drops, the spare lamp shall be installed in the otoscope head. The otoscope head shall be examined to determine how the light source is projected into the speculum. The light source shall be projected slightly below the aperture of the largest

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aperture speculum when viewed from the examiners side of the speculum. The illumination from the speculum shall be focused onto a flat, white surface, and the circle of illumination shall be examined for circularity and uniformity of illumination. The projected light from the otoscope head shall be compared with the illumination previously produced (4.4.1) by the instrument before being subjected to the drop test. The instrument shall meet the requirements specified in 3.3.2.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (eee 6.2).

5.1.1 Unit of issue. Each otoscope, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each otoscope consisting of items (a) thru (e) as specified in 3.2 shall be positioned within their respective compartments in the carrying case. The carrying case shall be closed and further packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure of the box shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Each otoscope shall be packaged in the carrying case as specified in 5.1.2.1. The carrying case shall be further packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Eighteen unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Eighteen unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged otoscopes shall be packed in shipping containers in a manner that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. These packs shall conform to the carrier's rules and regulations.

5.3 Marking. Unit packages and shipping containers shall be marked as specified in MIL-STD-129.

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6. NOTES

6.1 Intended use. The otoscope covered by this specification is intended for use in the examination of, and surgery within, the ear.

6.2 Ordering date. Procurement documents should specify the following:

a) Title, number and date of this specification.
b) National stock number (NSN).
c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.3 This specification covers the following item(s) appearing in the Federal Supply Catalog:

National stock number	Item identification
6515-00-550-7200	OTOSCOPE, BATTERY TYPE

6.4 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to previous issues, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA-DM

Project No. 6515-2121

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